Embodied First Exam

Coding Agents' "First Exam" for the Physical World

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1,156
tasks

47
robot types

35
suites

3
eval settings

Northwestern • CMU • UC Berkeley • Stanford • UW • UNC-Chapel Hill • Texas A&M • UT Austin

TL;DR: Strong with simulator “cheat access,” but ~30% under real-robot conditions

EFE evaluates coding agents on 1,156 tasks and 47 robots, letting them **choose their control method**. Top agent GPT-6 Astra solves **77%** of core tasks with full sim state and unlimited resets, but only **30%** with sensors only and no resets, as on real robots. The gap stems mainly from no retries and precision steps.

77% → 30%

GPT-6 Astra: Privileged → Standard
(391 core tasks with both settings)

33.6%

Top agent’s success rate on 456 core Standard tests

+25 vs +17

Gains from restoring “unlimited resets” vs. “exact state”
(pts; baseline 28%)

12 pts

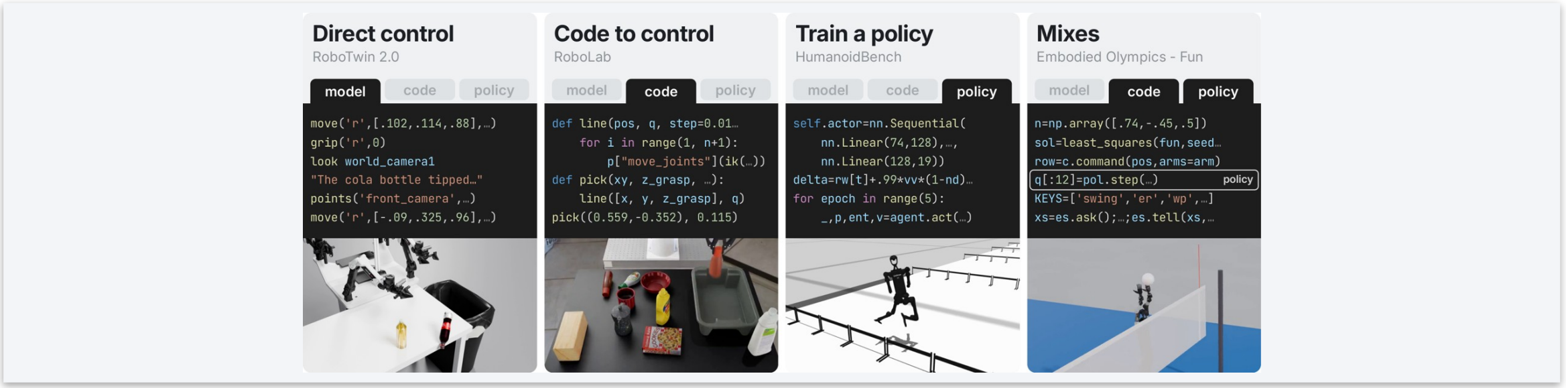
Max gap for one model across harnesses
(GPT-6 Astra: Codex 34% vs Claude Code 22%)

Motivation: Coding Agents May Be a “Third Path” to Robot Skills

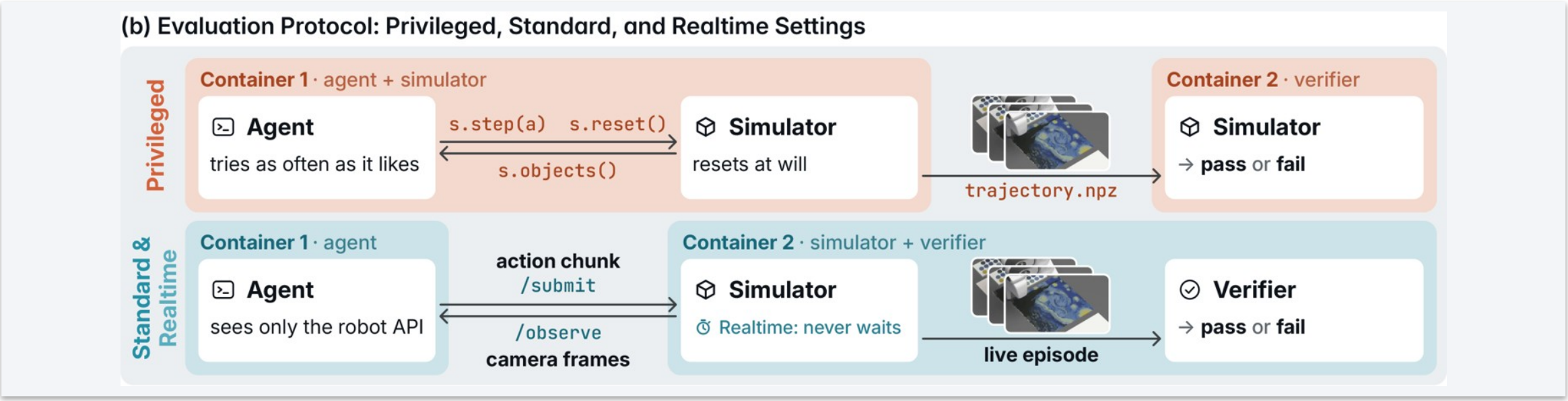
- Existing routes: **manual programming**, **learning from data** (VLA, world action models)
- New route: coding agents turn instructions into code and refine it via automated experiments
- Gap: robot benchmarks only evaluate visuomotor policies; coding benchmarks lack physical constraints

Core Question

Which robot tasks can general coding agents solve? Under what physical constraints? Where do they fail?



Protocol: Same Tasks, Three Physical Constraint Settings, Paired Comparison



Privileged

Near upper bound • yields training data

Shares sim container; reads full state
Unlimited resets, repeated trial & error
Isolated verifier replays & scores trajectory

Standard

Close to real robots

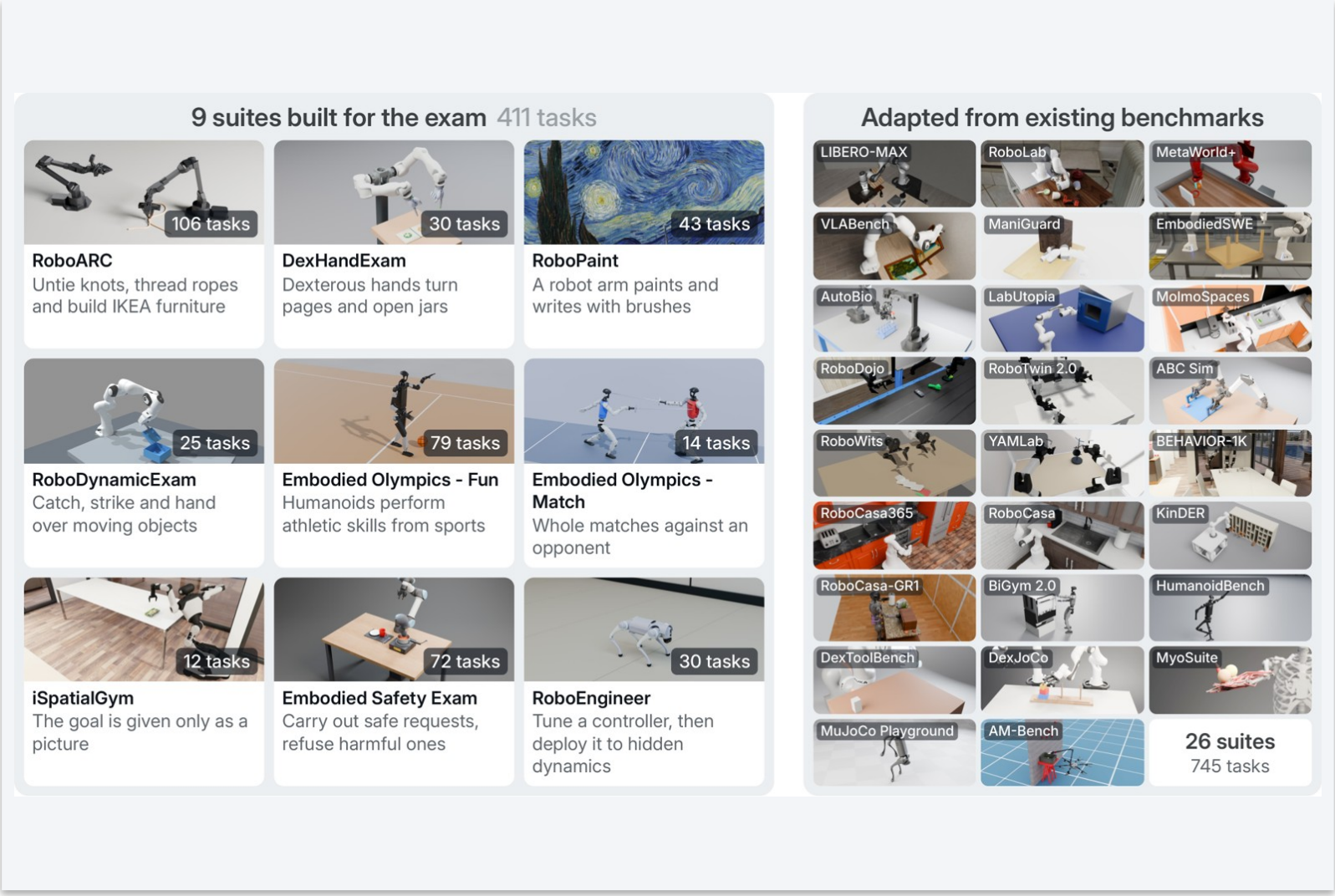
Proprioception + calibrated RGB-D only
No resets by default (3–50 for some tasks)
Sim pauses between requests; online scoring

Realtime

The world doesn’ t wait

Same interface as Standard
Sim never pauses (costly tasks slowed by fixed factor)
Dynamic tasks only, for now

Exam Content: 35 Suites, 47 Robots; 9 New Suites Go Beyond Pick-and-Place



35

suites: 9 new (411 tasks)
+ 26 adapted (745 tasks)

1,156

tasks / 2,288 tests

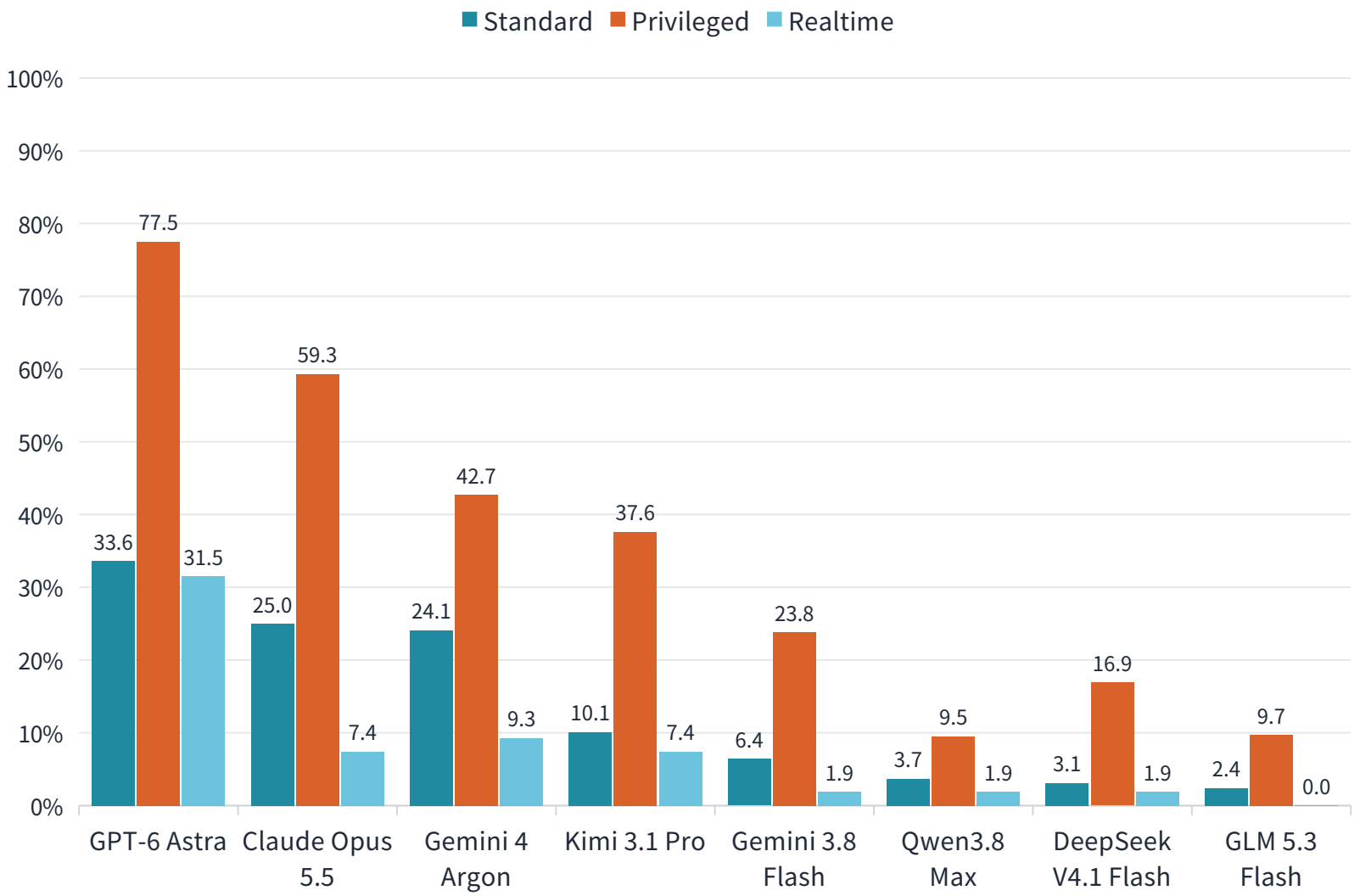
456

hard core tasks / 901 core tests

9

embodiments: single-arm, dual-arm, mobile manip., humanoid,
dex. hand, quadruped, musculoskeletal, aerial, wheeled

Overall: Even the Top Agent, GPT-6 Astra, Solves Only ~1/3 Under Standard



Top 3 > 2× rest

Top-3 Standard success (24–34%) is over twice that of the other five (⩽10%)

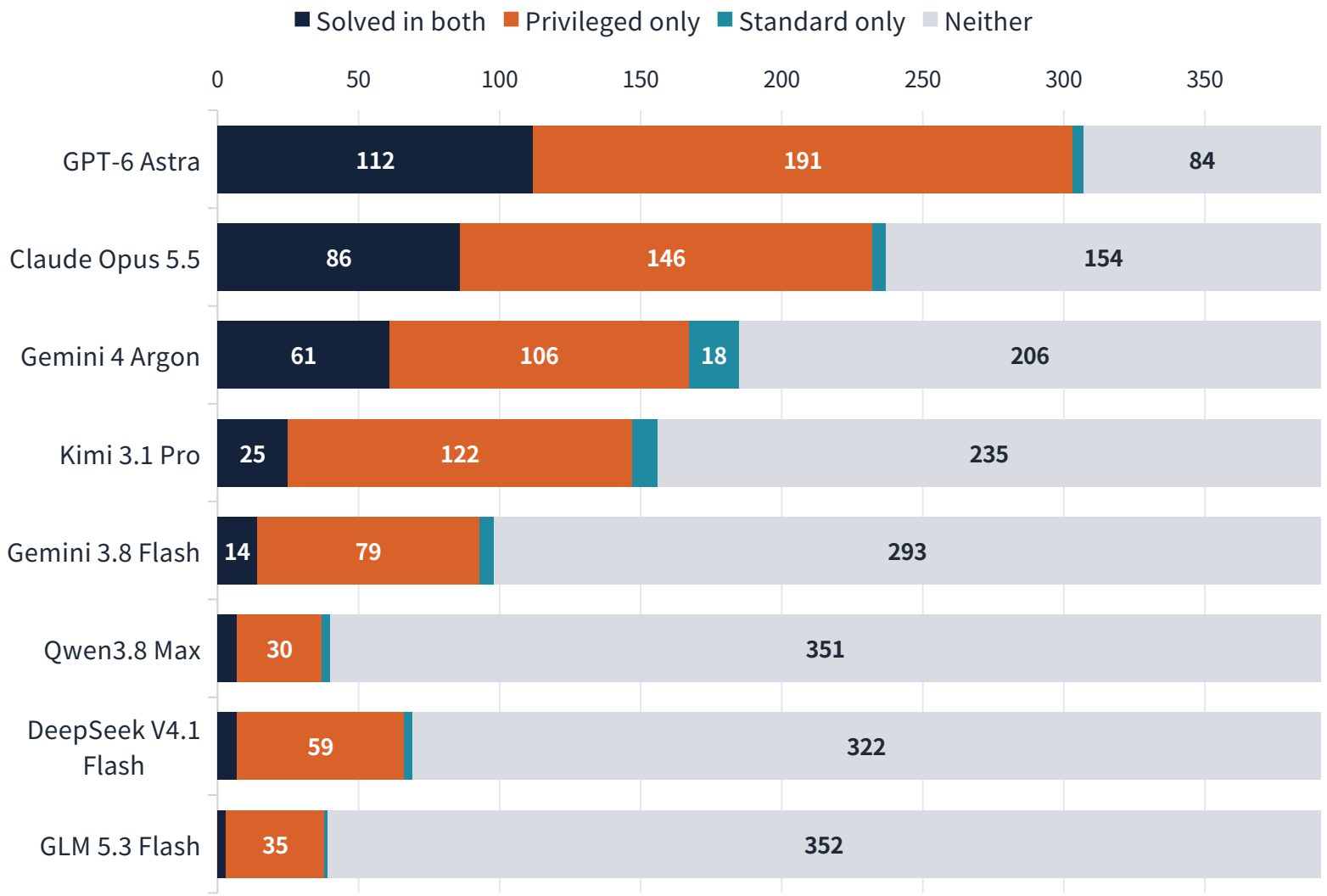
48%

Share of Standard tests solved by the top 3 combined

63% vs 50%

Unsolved by all, among 54 tasks with both settings:
Realtime vs Standard tests

Tasks Solved Under Privileged Mostly Fail Under Standard



303 → 112

GPT-6 Astra solves 303 in Privileged; only 112 of them also in Standard

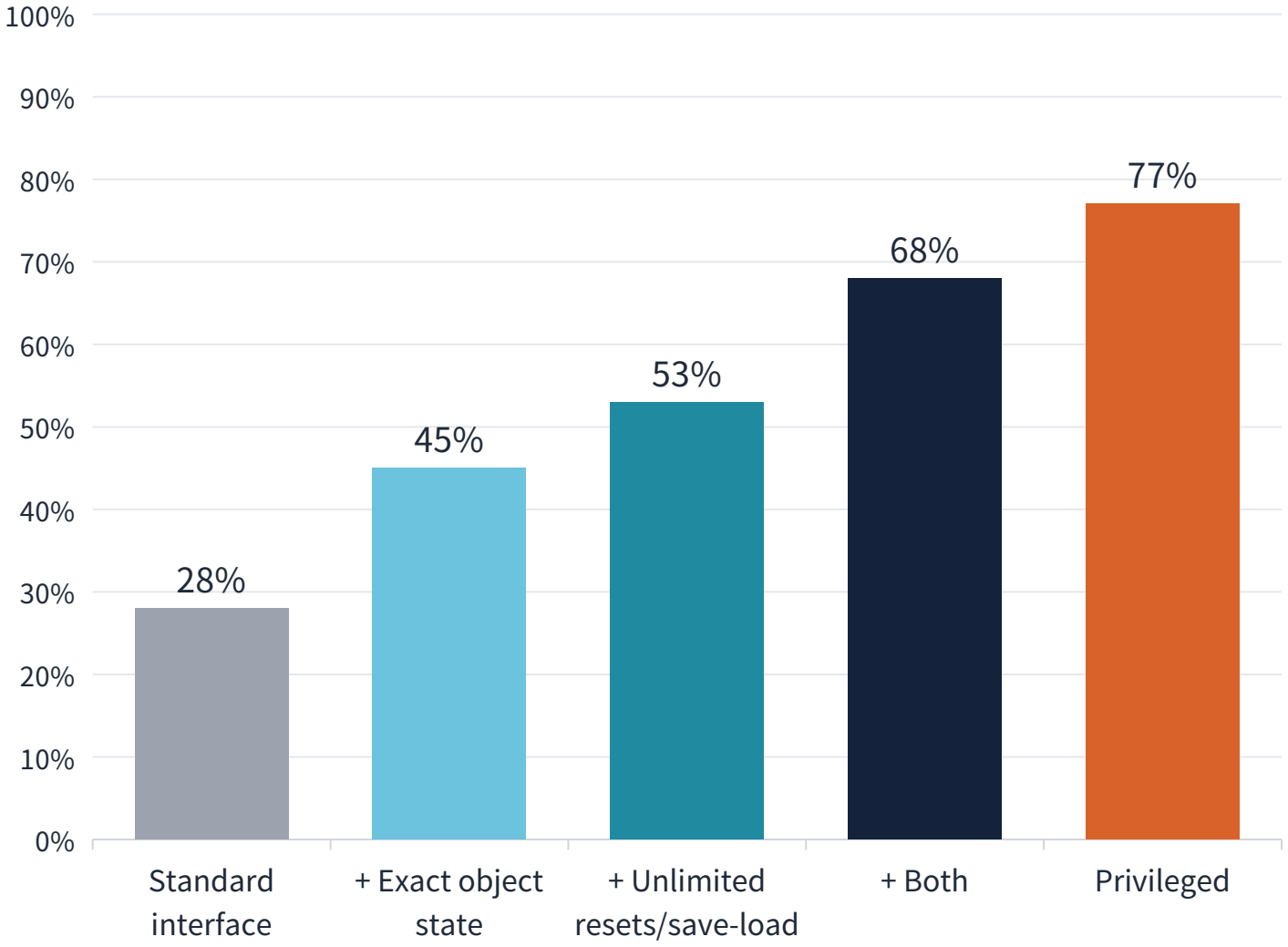
Only 4

Tasks GPT-6 Astra solves only in Standard

Bound ≠ per-task bound

Gemini 4 Argon solves 18 tasks only in Standard:
Privileged is an upper bound only at the task-set level

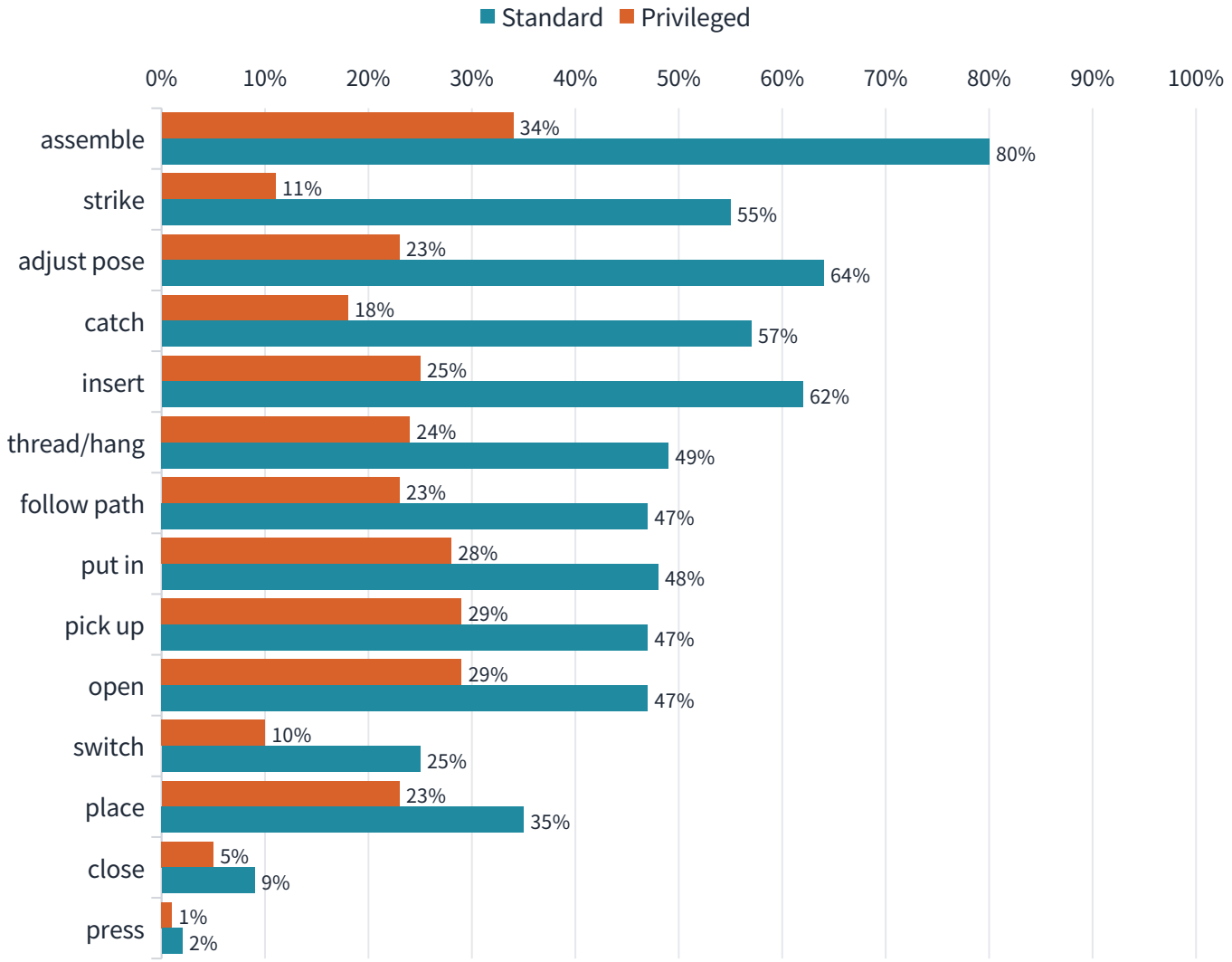
Restoring Unlimited Resets Narrows the Gap More Than Restoring Exact State



Key Numbers

- Resets: 28% → **53%** (+25)
- Exact state: 28% → **45%** (+17)
- Both combined: **68%**, near Privileged’ s 77%
- With resets, **66%** of successes come from episodes started at a save point; only 14% from the first episode
- Agents roll back to just before the hard step and retry, rather than restarting

Precision-Demanding Stages See Sharply Higher Failure Rates Under Standard



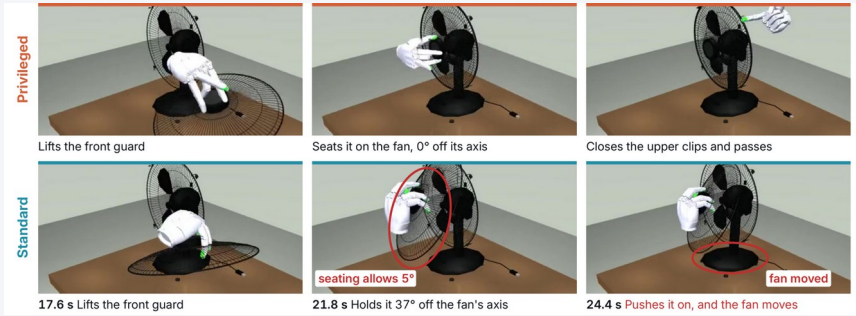
+24~46

Precision stages: failure-rate rise (pts)

+1~20

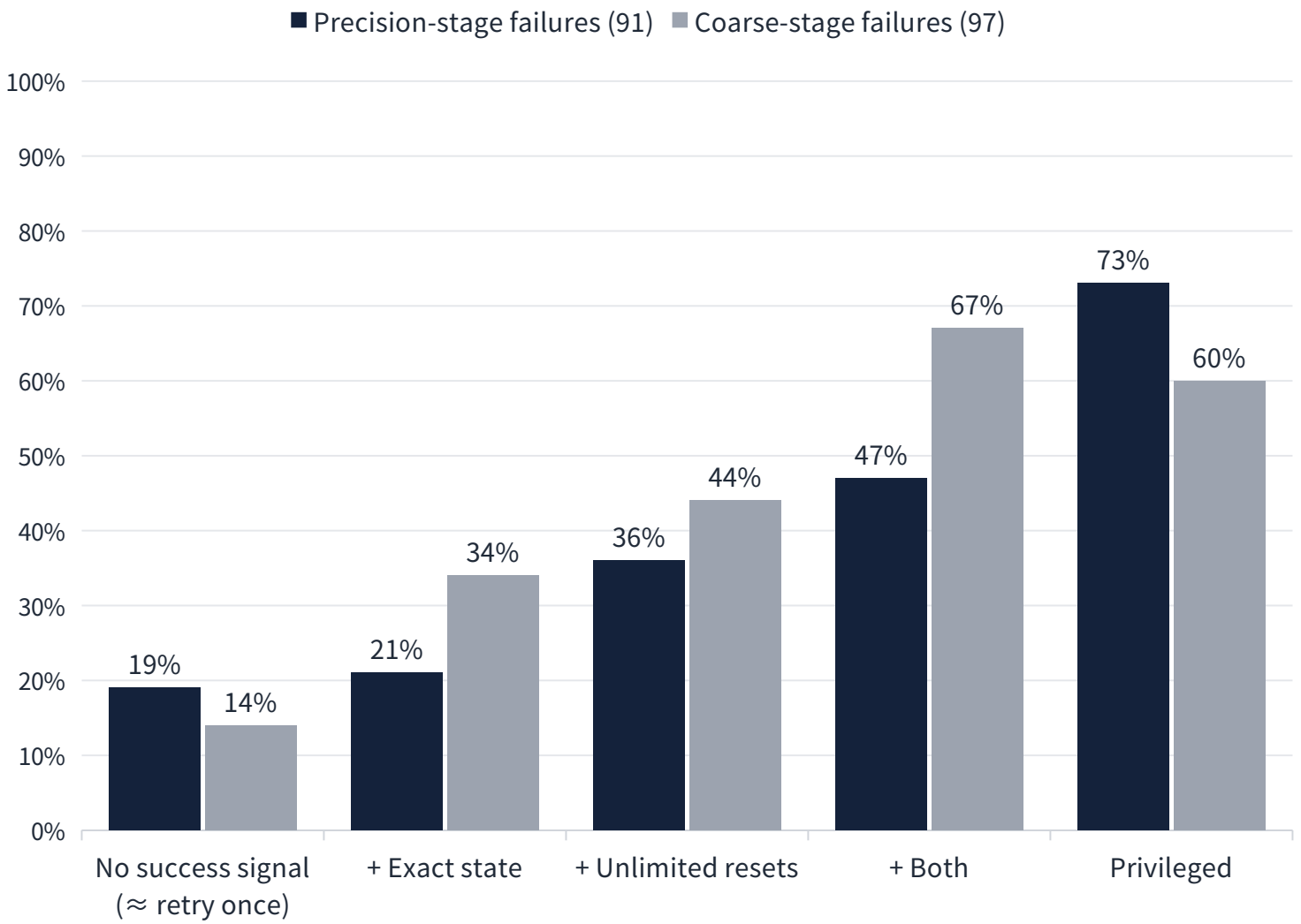
Coarse stages: failure-rate rise (pts)

Left: first 7 bars are precision stages, last 7 coarse stages



Claude Opus 5.5 mounting a fan front cover (DexHandExam): under Standard, the cover is off-axis by **36–39°** (5° allowed); one push shoves the whole fan away.

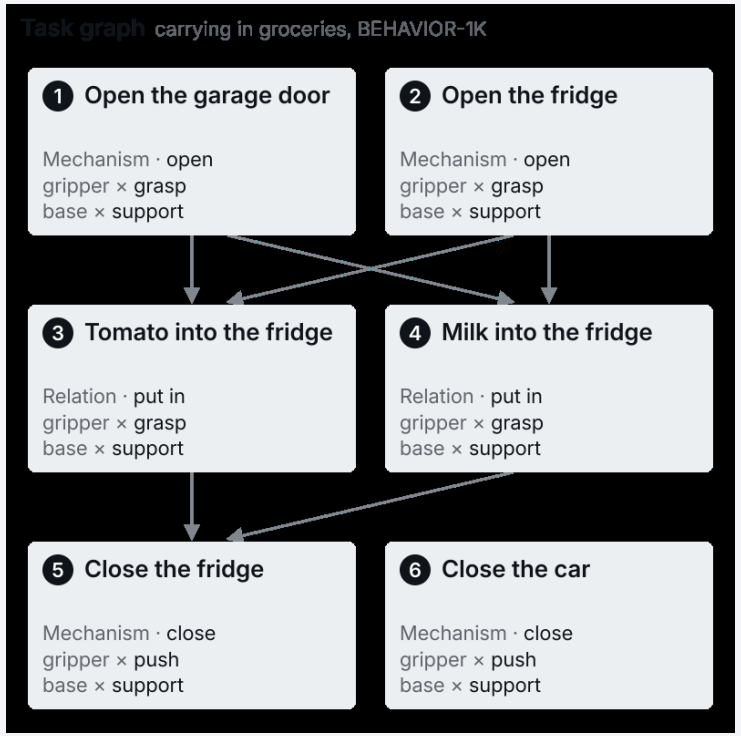
In Precision Stages, Exact State Alone Barely Helps; Resets Do



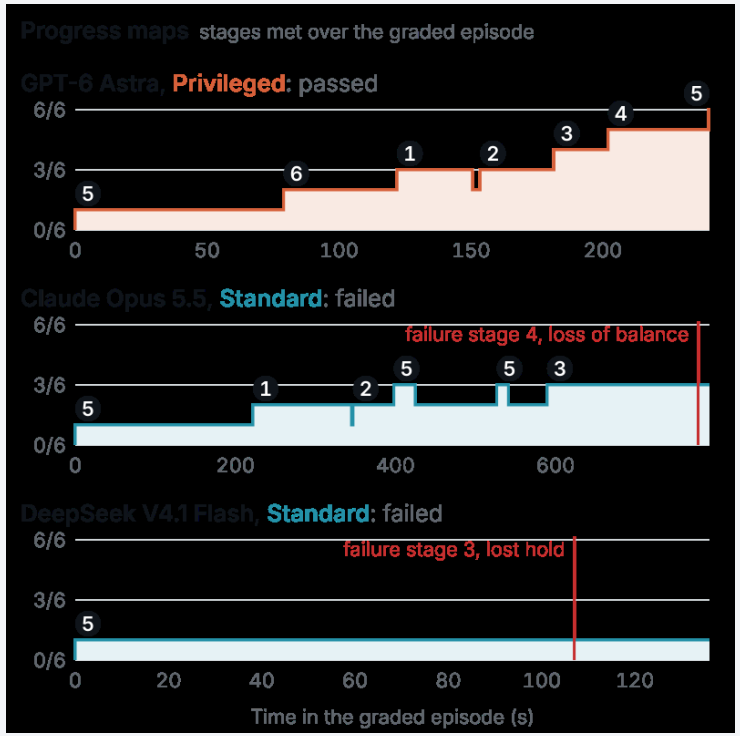
Takeaways

- Precision: state **21%** ≈ rerun 19%; resets raise it to **36%**
- Of 33 solved with resets, **31** came after a reset or load
- Coarse: state alone lifts 14% to **34%**
- Both added: precision 47% vs Privileged 73%; remaining gap mostly in follow-path, threading, assembly

Diagnosis: Decompose Tasks into Stage Graphs to Locate Each Failure



(a) Task graph: 6 stages; arrows denote required ordering



(b) Progress graphs, top to bottom: GPT-6 Astra (Privileged, pass) / Claude Opus 5.5 (Standard, loses balance at stage 4) / DeepSeek V4.1 Flash (Standard, loses grasp at stage 3)

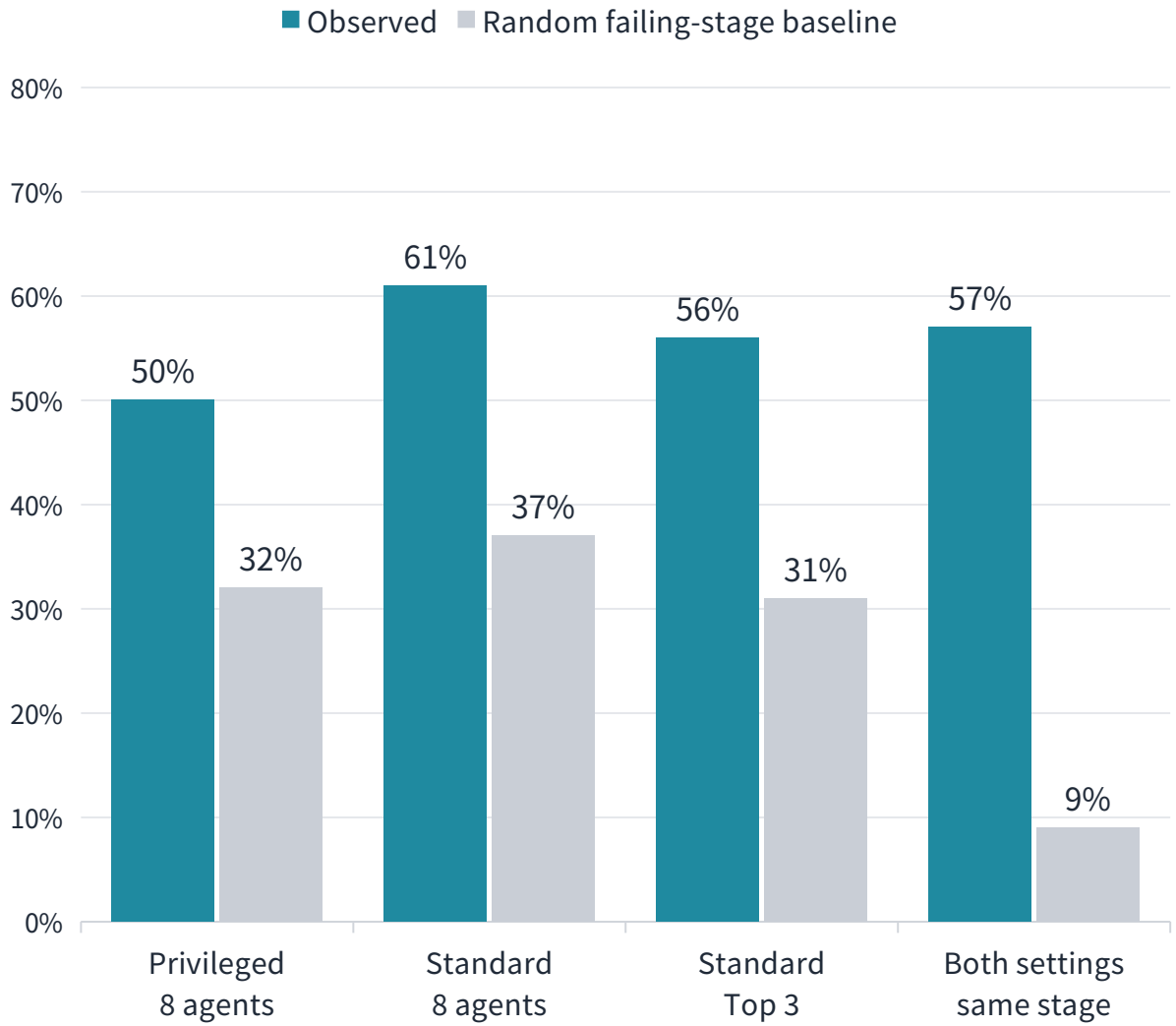
427
tasks decomposed into stage graphs

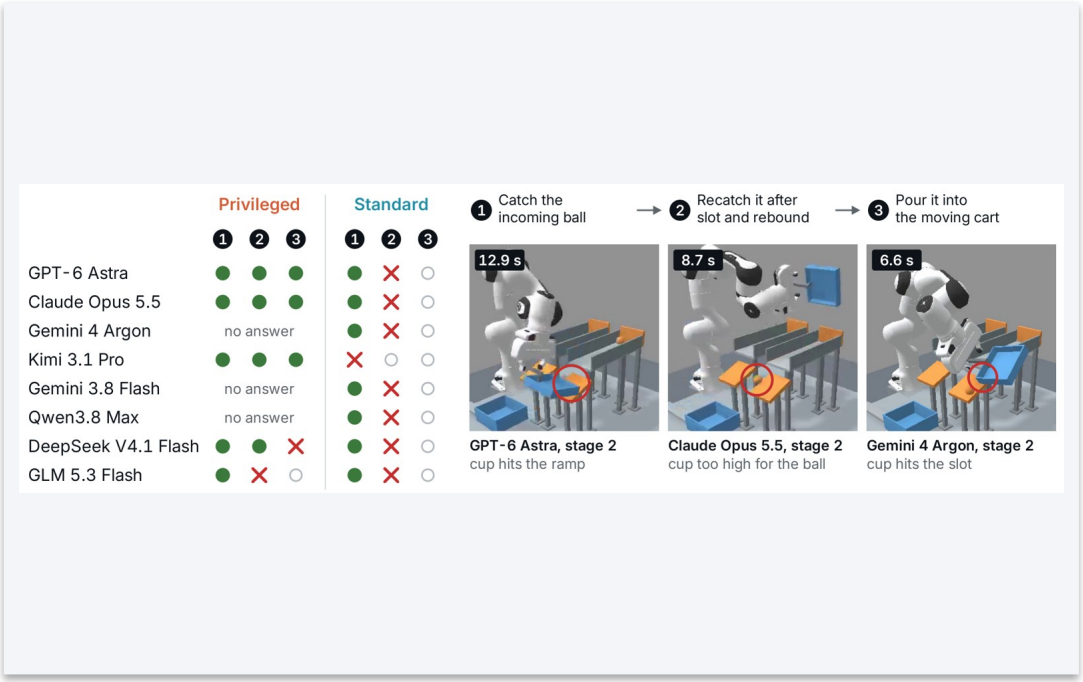
2,411
stages, avg. 5.6 per task

9
stage categories (relation, pose, dynamic, mechanism…)

6,779
attempts annotated

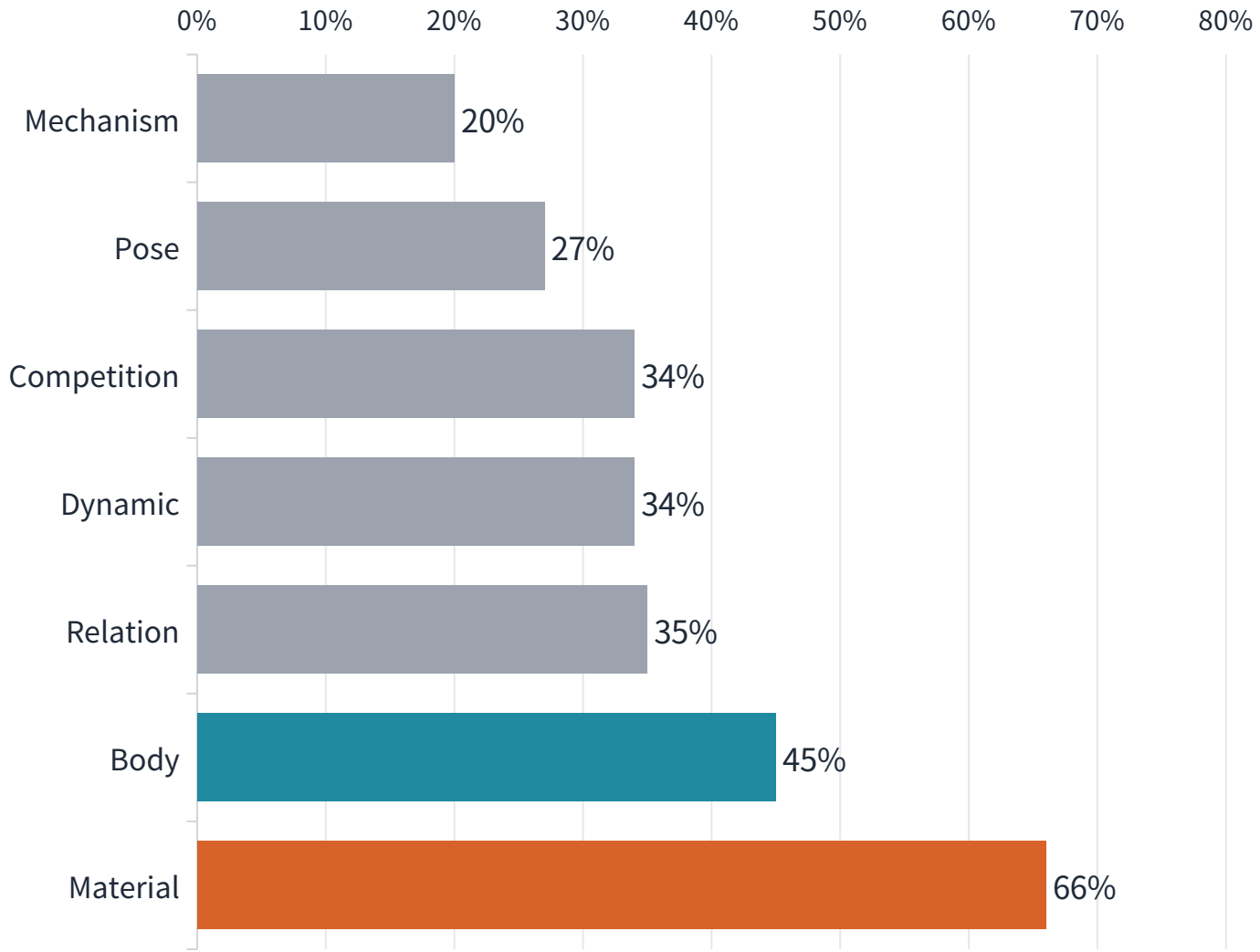
Failures Are Highly Concentrated: Different Agents Often Get Stuck at the Same Stage





- The most common failing stage is usually **the first stage** (65% of 264 Standard tests)
- In the catch-twice task, **7/8** agents complete the first catch but fail the second, each in a different way

Stages That Change Materials or Move the Body Fail Most Often



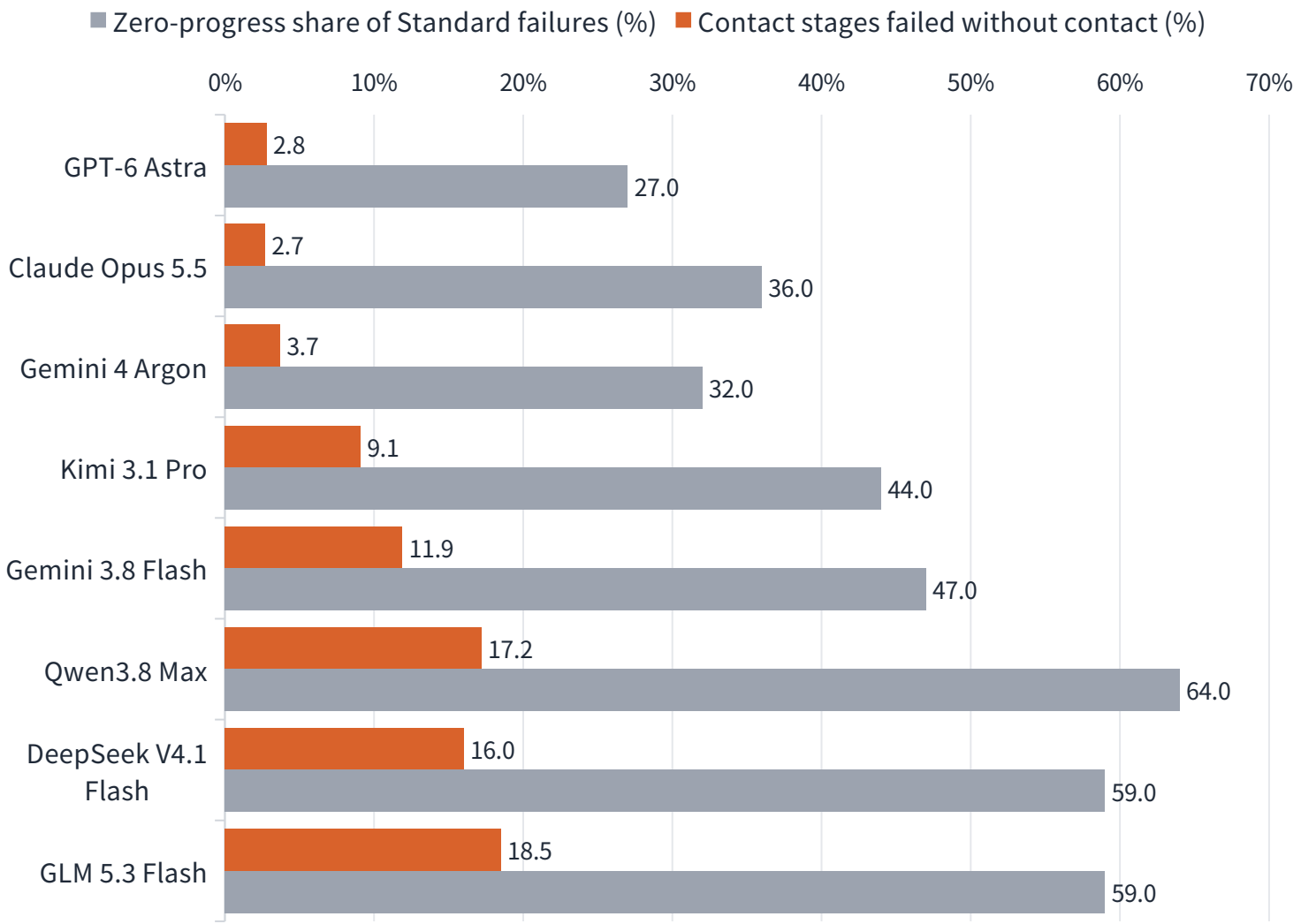
Large variation within categories

- Open **38%**, close 7%, press **3%**
- Relation stages: one hand 36%, bimanual 52%, legs/torso **65%**

Caution: correlation ≠ causation

Categories cluster in specific suites and robots. In leave-one-suite-out tests, only 55% of stage-type difficulty pairs keep their order, near chance.

Weaker Models More Often Fail “Without Even Touching the Object”



2.7–3.7%

Top 3: share of contact-requiring stages failed without any contact

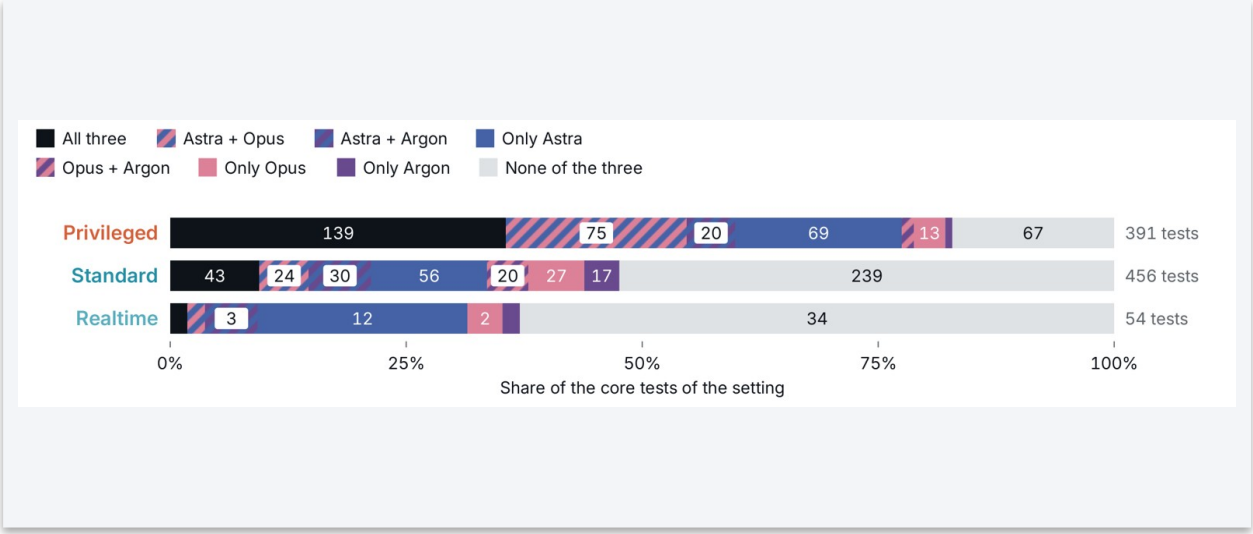
9.1–18.5%

Same metric for the five weaker models; higher in all 19 suites with enough samples

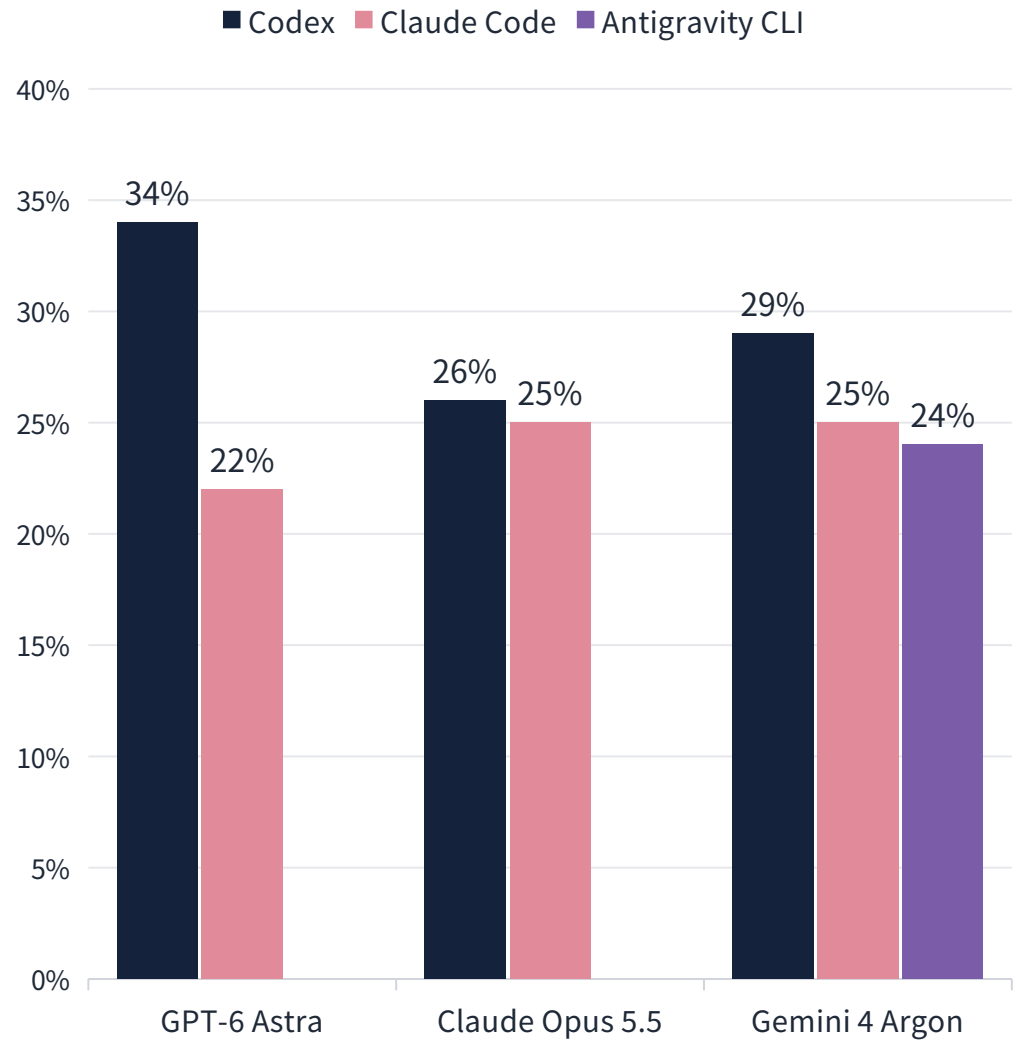
Normalize grasp loss by opportunity

GPT-6 Astra has the highest share of “grasp loss” failures (17%) because it reaches more grasp stages; per grasp stage, its rate is 5.6% vs 7.5% for other agents pooled (difference CI includes 0).

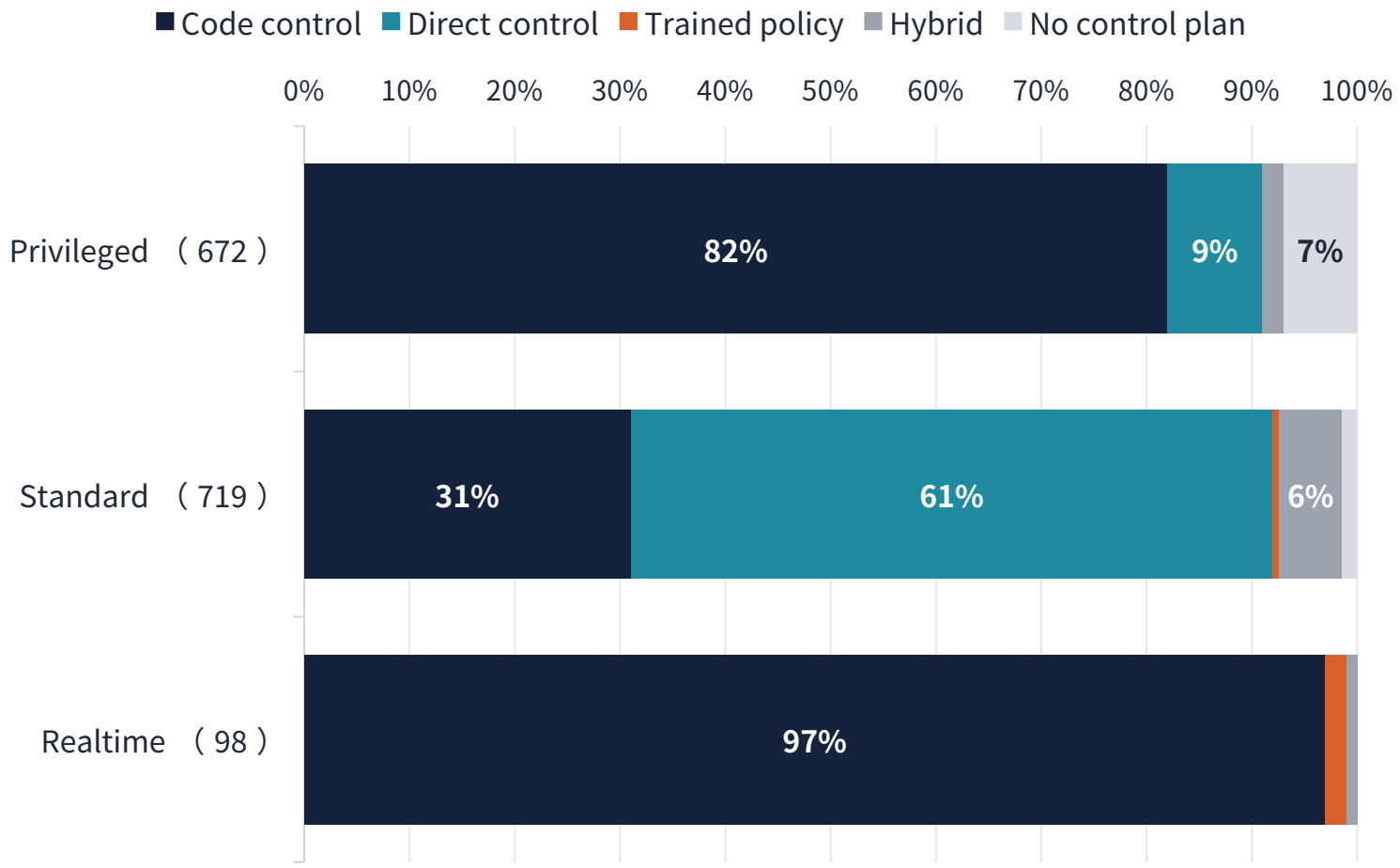
Top 3 Solve Partly Complementary Tasks; Harness Choice Shifts Results by up to 12 pts



- Standard: only Astra solves **56**, only Opus **27**, only Argon **17**, none of the three **239**
- Opus and Argon together solve **64** tests Astra fails; under Privileged, Astra covers 92%/95% of what they solve
- One run per task; some differences may reflect run-to-run variance



Control Shifts with Setting: Step-by-Step When Paused, Programs When the World Runs



Code-based control breakdown: Privileged closed-loop 45% / open-loop 20% / parameter search 17%; Realtime closed-loop 82%.

3 settings, 3 dominant modes

- **Standard 61%** direct control: model in the loop, viewing images after each action chunk
- **Privileged 82%** programs: compute or search actions from full state
- **Realtime 97%** programs; 82% run closed-loop in code
- E.g., in badminton, the program tracks the shuttlecock and fits a drag model before swinging; the agent edits it during pauses and wins **11–4**.

Soccer: With GPUs, Agents Choose to Train RL Policies; All Wins Come from Trained Policies

With a GPU

Trained walking and ball-control policies with RL

Won 10–8

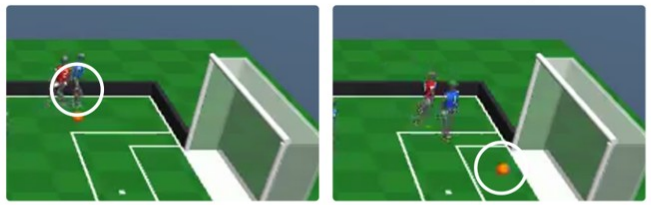
Robots walked 690 m (opponent 693 m)
Robots fell 22 times (opponent 10)

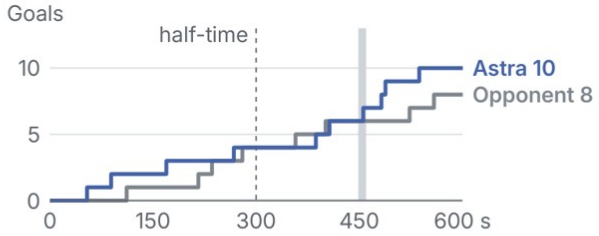
CPUs only

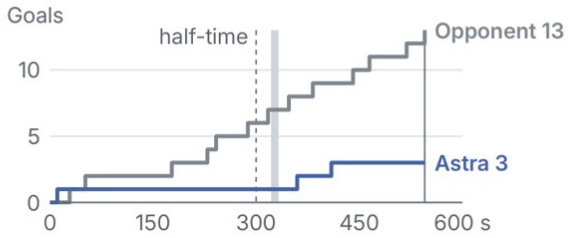
Hand-written gait tuned by search

Lost 3–13, mercy rule

Robots walked 276 m (opponent 624 m)
Robots fell 59 times (opponent 7)







17 / 20

GPU-version attempts trained RL policies (CPU version: 0 / 20)

4 / 20

GPU-version wins, all with trained policies (CPU version: 0 / 20)

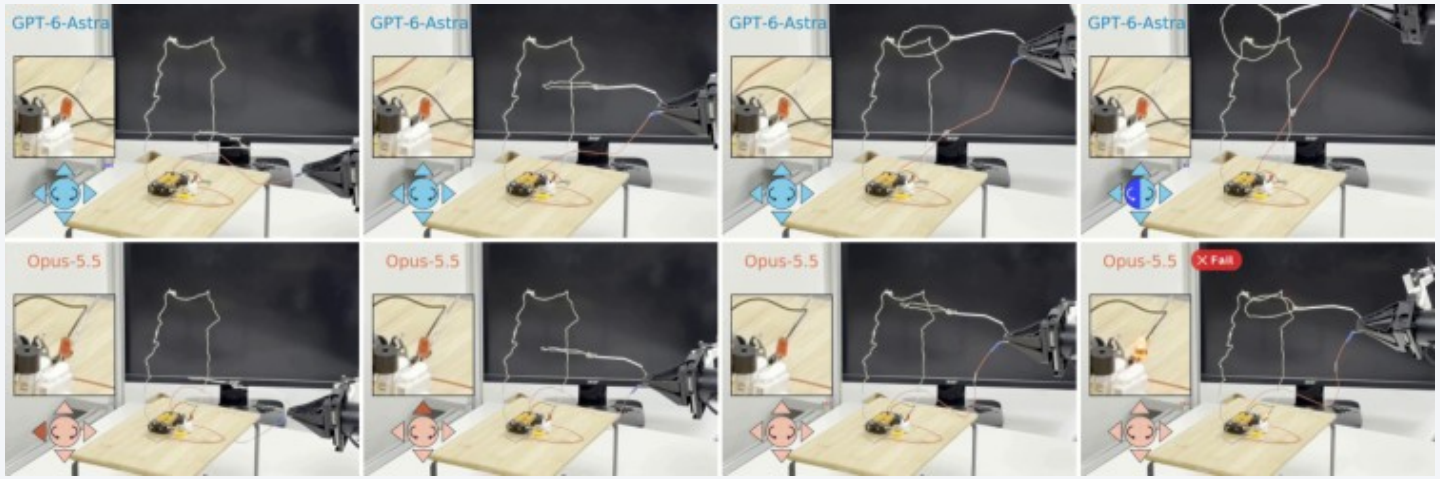
10–8 vs 3–13

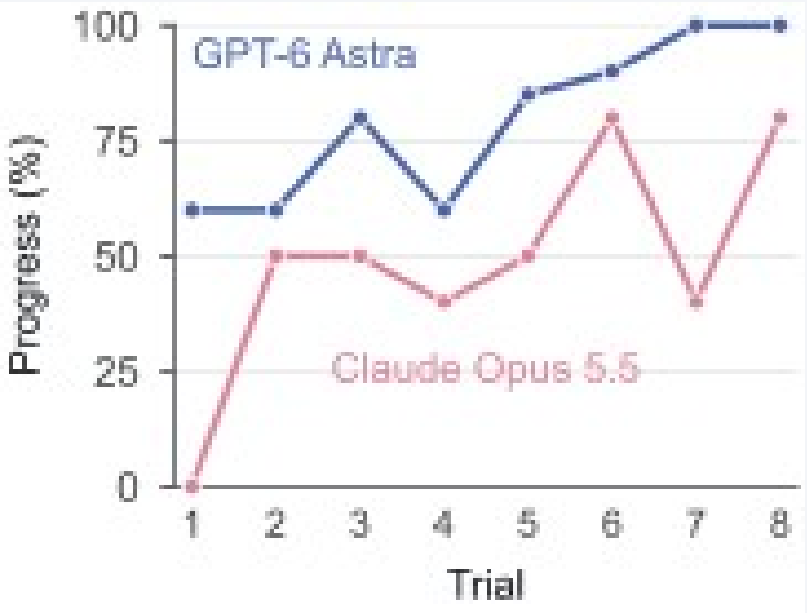
GPT-6 Astra: GPU version wins vs CPU version terminated early

Caution

Single task only; the two versions differ in GPU, RL libraries, training replicas and instruction examples, so the causal factor is unclear. CPU versions also won under Privileged, but every win reused the simulator's built-in opponent walking network.

Physical Robot Demo: The Standard Interface Connects Directly to a Real Arm





65.0% → 93.8%

GPT-6 Astra: mean progress, first 4 → last 4 trials

35.0% → 62.5%

Claude Opus 5.5: mean progress, first 4 → last 4 trials

Demo only: 1 task, one set of 8 trials per agent; not scored, and sim-to-real predictive validity is unverified.

Limitations: Single Attempts, Agent-Generated Labels, Simulation Only

Single attempt

Each test runs once, so ability can’ t be separated from run-to-run variance; part of the agents’ “complementarity” may be chance

Selection bias

Core set excludes tasks solved in GPT-6 Luna’ s pre-screen; Luna is in the same family as GPT-6 Astra

Agent-generated labels

Main annotator GPT-6.1 Sol shares GPT-6 Astra’ s family; labels can’ t separate perception from control errors; 47% of root causes are labeled only “Control”

Correlation ≠ causation

Stage categories are confounded with suites and robots; soccer findings come from one task, with versions differing in several ways

Simulation only

All in simulation except one physical demo; 392 tasks lack reference solutions

Possible memorization

For tasks adapted from public benchmarks, models may partly recall published solutions

Summary and Implications

Key Findings

1 Privileged 77% vs Standard 30%: restoring resets narrows the gap more than restoring exact state

2 Failures cluster at the same stage, usually the first; precision stages show the largest failure-rate rise under Standard

3 Weak models often fail to even make contact; top-3 solves are partly complementary; harness causes up to a 12-pt gap

4 Mostly direct control under Standard, programs under Privileged and Realtime; agents train RL policies when GPUs are available

Implications

- **Agent development**: save/rollback, closed-loop checks and delegating real-time control to programs may matter more than better perception alone
- **Robot learning**: "agent + trained policy" shows promise for dynamic whole-body control; treat resource allocation as an experimental variable
- **Evaluation design**: report results by "model × harness × resources"; the Standard interface transfers to physical robots
- **Future work**: multiple runs for confidence intervals, more Realtime tasks, systematic evaluation on physical robots